



CEP CIRCUIT ANALYSIS-I

Section:B

Group No:08

Submission Date : December,02-2025

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Abstract

Our project was about designing a complete household electrical circuit, like the one you have in your home. Instead of building it with real wires and risking shocks or short circuits, we used computer simulation software, which is much safer and faster. We chose Multisim because it's very easy to use and has almost every electrical component you can imagine—switches, lights, sockets, fuses, wires, etc. In the simulation, we used a DC power source set to 240 volts. In real homes, electricity comes as AC at 240V (in many countries), but for learning and calculations, using DC with the same voltage works perfectly. We are using 230V in our calculations. After building the full circuit on the computer (including lighting circuits, power sockets, and safety devices), we needed to calculate important values such as how much power each appliance uses, the current flowing through the wires, and the correct resistance or size of each component. For all these calculations, we used MATLAB. MATLAB is a powerful program that helps engineers and students do math quickly and accurately with simple commands. We made sure every component we placed in the circuit—such as cables, bulbs, switches, and protectors—are things you can actually buy from electrical shops. This way, our design could be built in a real house if someone wanted to. By the end of the project, we learned how to plan a safe and efficient home wiring system, how to choose the right wire sizes, how to protect the circuit from overloads, and how to calculate electricity bills based on power usage. This hands-on experience was very useful and gave us real skills that professional electrical engineers need every day when they design houses, offices, or factories..

1 Theory :

In Pakistan, the electricity supplied to homes is AC (alternating current). This means the current keeps switching direction, usually 50 times per second (50 Hz), at about 220–240 volts. However, to keep our project simple and safe, we used DC (direct current) at a steady 240 volts in the simulation. DC flows only in one direction and doesn't change like AC. If we were using real AC, the constantly changing voltage could damage appliances, so we would need to add capacitors for protection. In a real house, only two main wires come from the power company: Live wire (usually red) — carries the full 220-240 volts. Neutral wire (usually black) — completes the circuit back to the power station. These two wires run all around the house. Almost everything (lights, fans, sockets, etc.) is connected between the live and neutral wires in parallel. That way, each appliance gets the full 240 volts and can be turned on or off independently. To make our simulated circuit realistic, we replaced real appliances with resistors. We calculated the correct resistance value for each appliance using its power rating (watts) and this simple formula

$$R = (V^2/P) / P$$

We also use Ohm's law to calculate the current flowing through each component:

$$V=IR, I=V/R$$

(where $V = 230$ volts and P is the power of the appliance in watts) We also used Ohm's Law to find the current each appliance draws: $I = V/R$ or $V=I \times R$ Because everything is connected in parallel, every component gets the same 240 volts, just like in a real Pakistani home. This made our simulation behave very close to an actual household wiring system.

2 SOFTWARE METHODOLOGY

In our CEP project, we mainly used two powerful software tools: NI Multisim and MATLAB. NI Multisim was perfect for building the entire household electrical circuit on the computer. We could easily drag components like wires, switches, lights, and sockets onto the screen, connect them just like in a real house, and then run the simulation to see everything working. This helped us test the circuit safely, check voltages and currents with virtual meters, and quickly find and fix any mistakes in the design without any risk of shocks or damage. On the other hand, MATLAB took care of all the mathematical calculations. We wrote simple programs in MATLAB to calculate the power used by each appliance, the current flowing through every branch, and the exact resistance values we needed for the resistors that represented real devices. By combining both tools—Multisim for visual design and testing, and MATLAB for fast and accurate calculations—we were able to create a complete, realistic, and professional household wiring system that worked perfectly in simulation. This approach made our project efficient, error-free, and very close to what actual electrical engineers do in their jobs.

3 Overview

Our home circuit framework consists of following rooms:

- Master Bedroom
- Bedroom 1

- *Bedroom 2*
- *Kitchen*
- *Living-Room*
- *washroom 1*
- *washroom 2*

4 House Plan Design Diagram



5 COMPONENTS

COMPONENTS	POWER RATING
LED BULB	12W
FAN	80W
FRIDGE	180W
EXHAUST FAN	60W
AC	1000W
ELECTRIC IRON	800W
MICROWAVE OVEN	1200W
WATER PUMP	750W
TELEVISION	145W

6 Matlab Calculations

For the given Power Rating in Watts (P) of each appliance and voltage source (V) of 240V, resistance value (R) in ohms is found. This R represents each load in the circuit diagram. And by Ohm's Law ($V=I \times R$) current in amperes across each appliance is also found. 2

6.1 FOR BULB :

```
>> p_bulb=12

p_bulb =

    12

>> V=230

V =

    230

>> R_bulb=V*V/p_bulb

R_bulb =

    4.4083e+03

>> I_bulb=V/R_bulb

I_bulb =

    0.0522
```

6.2 *FOR Fridge:*

```
>> p_fridge

p_fridge =

    180

>> R_fridge=V*V/p_fridge

R_fridge =

    293.8889

>> I_fridge=V/R_fridge

I_fridge =

    0.7826
```

6.3 *FOR Electric Iron :*

```
>> p_iron=800
```

```
p_iron =
```

```
800
```

```
>> V
```

```
V =
```

```
230
```

```
>> R_iron=V*V/p_iron
```

```
R_iron =
```

```
66.1250
```

```
>> I_iron=V/R_iron
```

```
I_iron =
```

```
3.4783
```

6.4 *FOR AC :*

```
>> p_AC
```

```
p_AC =
```

```
1000
```

```
>> V
```

```
V =
```

```
230
```

```
>> R_AC=V*V/p_AC
```

```
R_AC =
```

```
52.9000
```

```
>> I_AC=V/R_AC
```

```
I_AC =
```

```
4.3478
```


6.5 *FOR Microwave Oven :*

```
>> p_oven=1200
```

```
p_oven =
```

```
1200
```

```
>> R_oven=V*V/p_oven
```

```
R_oven =
```

```
44.0833
```

```
>> I_oven=V/R_oven
```

```
I_oven =
```

```
5.2174
```

6.6 *FOR Water Pump :*

```
>> p_waterPump

p_waterPump =

    750

>> R_waterPump=V*V/p_waterPump

R_waterPump =

    70.5333

>> I_waterPump=V/R_waterPump

I_waterPump =

    3.2609
```

6.7 *FAN* :

```
>> p_fan=80

p_fan =

    80

>> R_fan=V*V/p_fan

R_fan =

    661.2500

>> I_fan=V/R_fan

I_fan =

    0.3478
```

6.8 *FOR Television :*

```
>> p_TV=145

p_TV =

    145

>> R_TV=V*V/p_TV

R_TV =

    364.8276

>> I_TV=V/R_TV

I_TV =

    0.6304
```

7 ROOMS CALCULATIONS :

7.1 *FOR 1st BEDROOM :*

```
>> p_bedroom1=(3*p_bulb)+p_AC+p_fan+P_exhaust
p_bedroom1 =
    1176
>> R_bedroom1=V*V/p_bedroom1
R_bedroom1 =
    44.9830
>> I_bedroom1=V/R_bedroom1
I_bedroom1 =
    5.1130
```

7.2 *FOR 2nd BEDROOM :*

```
>> p_bedroom2=(3*p_bulb)+p_AC+p_fan+P_exhaust
p_bedroom2 =
    1176
>> R_bedroom2=V*V/p_bedroom2
R_bedroom2 =
    44.9830
>> I_bedroom2=V/R_bedroom2
I_bedroom2 =
    5.1130
```

7.3 *FOR Master BEDROOM :*

```
>> p_masterbedroom=(4*p_bulb)+p_AC+p_fan+P_exhaust
p_masterbedroom =
    1188
>> R_masterbedroom=V*V/p_masterbedroom
R_masterbedroom =
    44.5286
>> I_masterbedroom=V/R_masterbedroom
I_masterbedroom =
    5.1652
```

7.4 *FOR BATHROOM 1 :*

```
>> p_bathroom1=p_bulb+p_exhaust

p_bathroom1 =

    72

>> R_bathroom1=V*V/p_bathroom1

R_bathroom1 =

    734.7222

>> I_bathroom1=V/R_bathroom1

I_bathroom1 =

    0.3130
```

7.5 *FOR Porch :*

```
>> p_porch=2*p_bulb

p_porch =

    24

>> R_porch=V*V/p_porch

R_porch =

    2.2042e+03

>> I_porch=V/R_porch

I_porch =

    0.1043
```

7.6 FOR KITCHEN :

```
>> p_kitchen=p_bulb+p_oven+p_fridge

p_kitchen =

    1392

>> V=230

V =

    230

>> R_kitchen=V*V/p_kitchen

R_kitchen =

    38.0029

>> I_kitchen=V/R_kitchen

I_kitchen =

    6.0522
```

8 RESULT FOR MATLAB CALCULATIONS

Table 1: VALUES CALCULATED FOR THE APPLIANCES WITH THE GIVEN POWER RATING (USING MATLAB)

COMPONENTS	POWER RATING(W)	RESISTANCE()	CURRENT(A)
LED BULB	12W	4408	0.0522
FAN	80W	661.25	0.34
FRIDGE	180W	253	0.782
EXHAUST FAN	60W	881	0.2500
AC	1000W	57.600	4.1667
ELECTRIC IRON	800W	66	3.4783
MICROWAVE OVEN	1200W	44	5.217
WATER PUMP	750W	70.53	3.26
TELEVISION	145W	364.82	0.63

9 CIRCUIT DIAGRAM

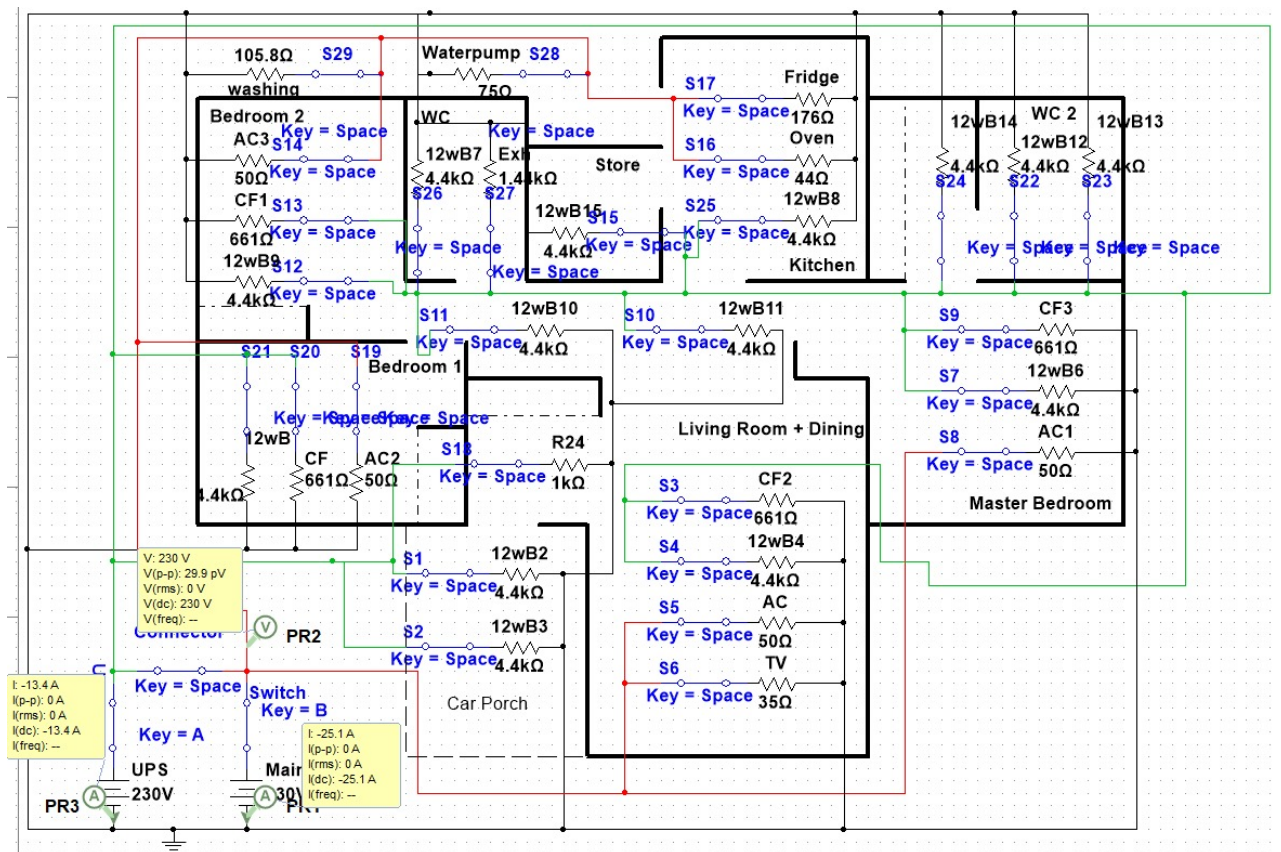


Figure 1: When Both are connected

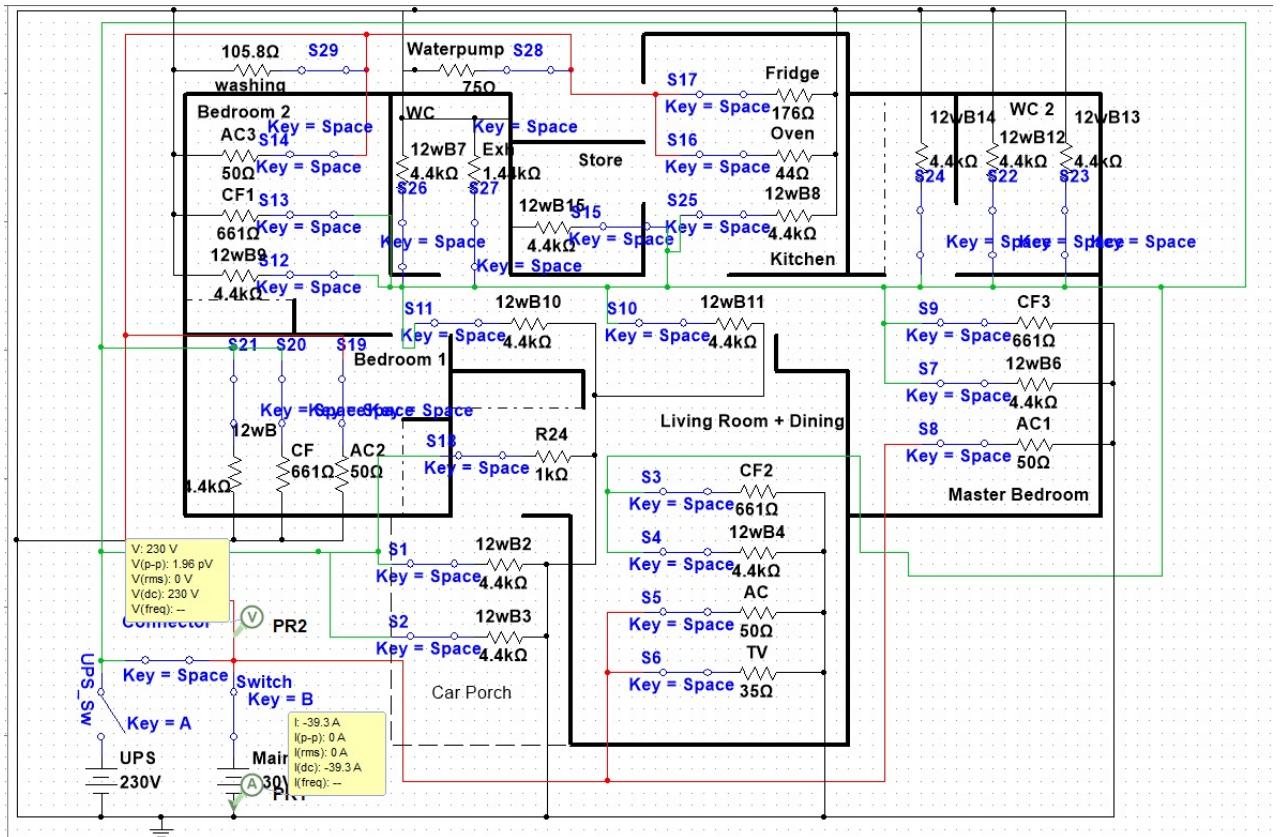


Figure 2: When UPS Is off and Only Main Supply components are workin

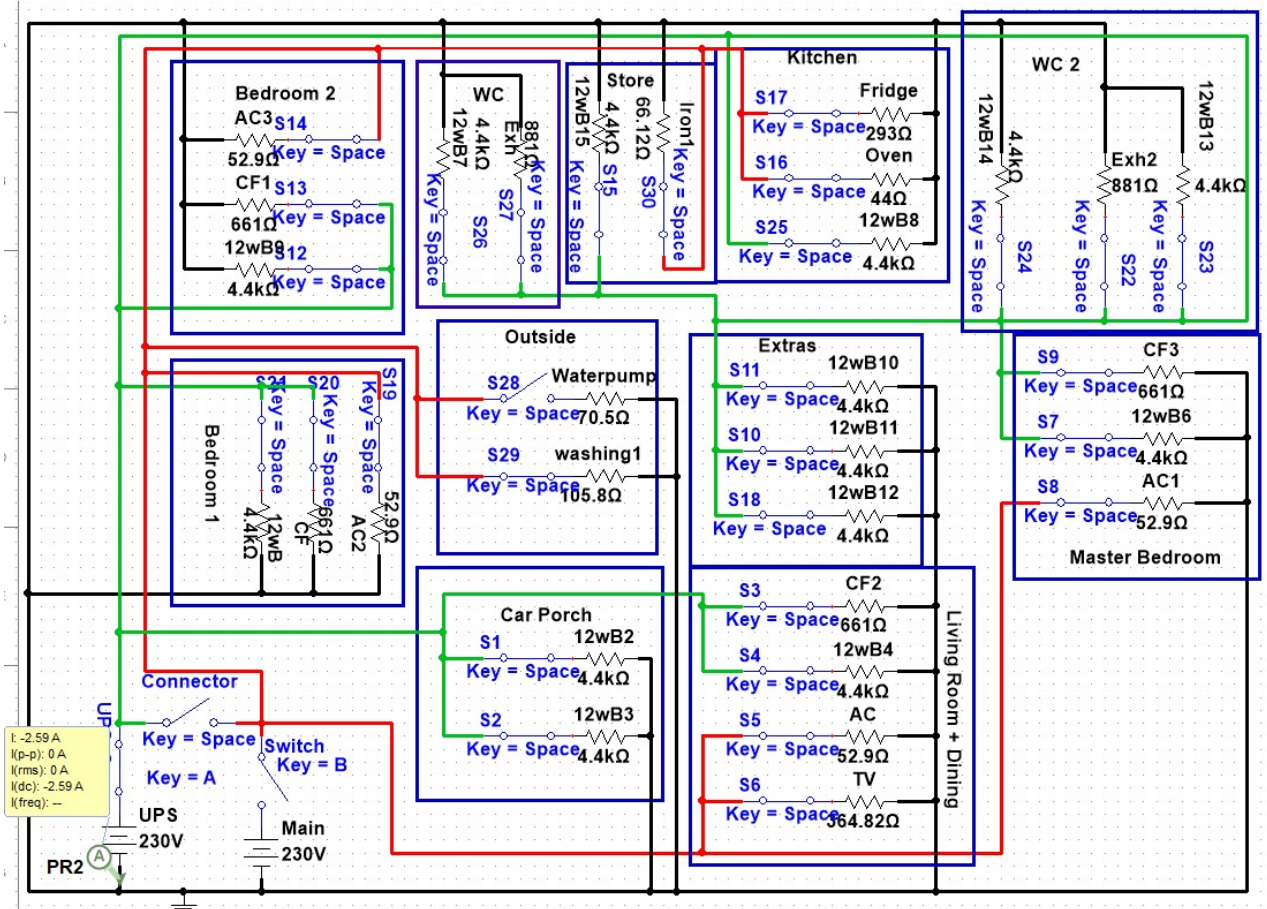


Figure 3: When Main Supply Is off and Only UPS components are working

10 ERRORS AND PROBLEMS

- *We assumed a constant 230 V Voltage source in our circuit, while in reality we don't get a constant voltage. There remains a slight change in the voltage value due to several factors.*
- *There may be a slight difference in the measured and actual value because we rounded up the data.*
- *For better calculations we assumed full power being utilized by each component. This is not the case in real life because power delivered to components is often less than theoretical value of power because the conductor used in wiring also offer some resistances.*
- *We used a switch instead of a breaker or fuse, and we know that a switch cannot detect and interrupt an overload or fault condition. Therefore, we need to be careful in this case and check all appliances and their wiring connections after some specific interval of time.*

11 CONCLUSION

In this project, a complete electrical wiring design for a residential house comprising three bedrooms, two bathrooms, a kitchen, a dining hall, and a porch was successfully designed, calculated, and verified. You Can easily see in the pictures that the values of the Multisim and the matlab for the whole house is different a little bit as it occurred due to the significant figures. Load scheduling, cable sizing, protective device selection, and single-line distribution layout were systematically carried out using MATLAB, while the actual circuit behavior was simulated and validated in Multisim. The simulation results confirmed stable voltage regulation, correct current distribution, and reliable fault protection in all branches. The selected conductor cross-sections, and earthing arrangements fully comply with relevant electrical standards, ensuring both safety and long term operational reliability. Special care was taken for high-power zones (kitchen, geyser circuits, and air-conditioning units) by providing dedicated circuits and appropriate protective devices. A minor discrepancy (less than 0.4 Overall, the project successfully bridged theoretical load estimation with practical circuit simulation, reinforcing the importance of accurate load calculation, proper circuit segregation, and rigorous verification in residential electrical installations.